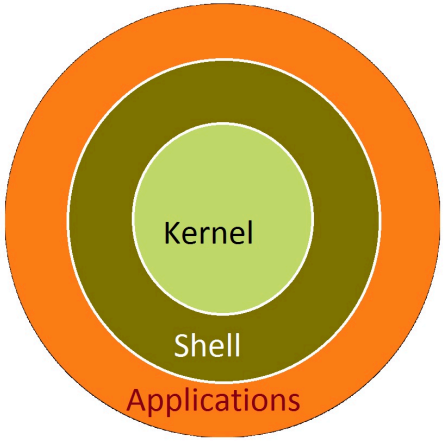
Introduction to Linux

==========

- What is a software
  - Software is a collection of individual programs and they are grouped together to perform a specific purpose.
  - Purpose that defines the category of the software
  - Categories
    - System Software : designed to manage the system or resources of the system. Also known as the operating system. Example : Windows and Linux
    - Application Software : ensures that we get the things we are expecting from the software. Example : VLC , Vmware , MS Office
    - Utility Software : helps both application and system software to work in a better way. Example : Antivirus , Disk defragmenter
- What is Linux :
  - Linux is an open source operating system.
- What is an operating System
  - An operating system is a software that manages all of the resources associated with the desktop,laptop or server system.
  - The operating system manages the communication between the software and the hardware.

- Without the operating system the software will not perform.

-
- Anatomy of an Operating System :
  - Kernel : core component/program of an operating system that interacts directly with the hardware. Uses Assembly language or machine level language(0,1). It works like a bridge between system hardware and other user defined applications.
  - Shell : it is a program which functions like an interpreter between the kernel and application as the kernel uses machine level language and application uses high-level human friendly languages.
  - Application : used by the user to provide instruction to the system to perform a specific task.



- Function of Kernel
  - Memory Management
  - Process Management
  - I/O Device Management
  - Security Management
- Who developed Linux Kernel
  - Linus Torvalds in 1991
  - A MCA 3rd Year student from helsinki university at the age of 21
  - It was a hobby
- What is open source software
  - Source code is the human readable instruction that is written to help the computer to perform a specific function from where the software gets built up.
  - Open source software is a software with source code that anyone can use , study , modify and share for free.
  - Some software has source code that only the person , team or organization that created it can see or change or distribute. These softwares are known as proprietary or closed source software.
  - Open source promotes collaboration , sharing , transparency and rapid innovation because it encourages people beyond the original developers to make modifications and improvements to the software and to share it with others.

- A community for open source is Open Source Community.

- Benefits of Open Source
  - Control - see what the code does and change it to improve it.
  - Training - Learn from real-world code and develop more useful application
  - Security - Inspect sensitive code, fix with or without the original developer help
  - Stability - Code can survive the loss of the original developer or distributor.

- License type of Open Sources :
  - Copyleft License :
    - Encourages keeping the source code open
    - Distribute it but keep it open for others.
    - Example : GNU General Public License(GPL) , Lightweight GNU General Public License (LGPL)
  - Permissive License :
    - Designed for maximise the reusability
    - Reuse the code but don't perform any modification.
    - These softwares are used as a platform on which other softwares can run
    - Example : Redhat uses a graphical framework called Wayland. On this platform redhat supports 2 graphical UI to run. GNOME and KDE.
    - GNOME : GNU Network Object Model Environment
    - KDE : K Desktop Environment

        - If anyone GNOME or KDE changes the wayland for their application , the other application will not run.
        - So it is recommended not to change the Product which is carrying a permissive license.
        - Example : MIT/X11 License , Simplified BSD License and Apache software license 2.0
- What is Linux Distribution
    - It is also known as Linux Distro
    - It is an operating system compiled from components developed by various open-source projects and programmers.
    - Each distribution includes linux kernel(For Hardware interaction) , the GNU Shell utilities(for terminal interface and command) , X Server (For graphical desktop) , the desktop environment , a package management system , an installer and other services.
    - Many components are developed independently from each other and are distributed in source code form by different developers.
    - Linux distributions compile the codes from open-source projects and combine ir into a single operating system which can be installed and booted up.
- Popular Linux distributions
    - Debian
    - Redhat

    - SUSE
- What is debian
    - It is a family
    - It includes one of the popular operating system called “Ubuntu”
    - The name debian came from the way the compilation is done called “deb”.
    - The applications are distributed in .deb format
    - Distributed by a UK Based organization called “Canonicals”
    - It is free to download , install and use.
    - Available editions are server , desktop and core.
- What is Redhat
    - First release in 2002
    - It is a server operating system
    - It is not free
    - It is available in various sorts of subscription based pricing model which are designed based on the end customer requirements.
    - Provides 3 main offers
        - RHEL
            - Is the flagship os
            - Used in servers in most of the organization
        - CentOS
            - Free version

            - Offers same packages
            - No support available
            - Supports are given from technical forums
        - Fedora
            - Redhat sponsored product
            - With new features
- What is SUSE
    - Not that popular
    - But considered as one of the mainstream linux distributions
    - Founded as a Unix type system in Germany in 1992.
    - Currently owned by EQT, a Swedish Private equity group
    - Provides 2 offers
        - OPENSUSE
            - Free
            - Provides latest updates
            - 2 editions
                - Tumbleweed : continuously updates itself
                - Leap : provides stable release every 6 months
        - SUSE Enterprise Linux (SLES)
            - Commercially priced
            - Highly available

- Server OS